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TITLE OF INVENTION

System and Method for Slice-Topology-Aware Causal Root Cause Analysis and Closed-Loop Remediation for Cloud-Native 5G Standalone Core Networks

Inventor:	Krishna Kumar Gattupalli
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1. TECHNICAL FIELD

The present invention relates to network management systems for fifth-generation (5G) standalone (SA) core networks. More specifically, the invention relates to real-time fault isolation and root cause analysis using causal directed acyclic graphs (DAGs) that are structurally aware of 5G network slice topology, Service-Based Interface (SBI) dependencies, and Packet Forwarding Control Protocol (PFCP) session bindings, enabling disambiguation between network function (NF)-layer faults and slice-layer faults in multi-slice cloud-native deployments. The invention further relates to methods, systems, and non-transitory computer-readable media for automated real-time fault isolation through dynamic causal graph construction from multi-modal telemetry streams.

2. BACKGROUND OF THE INVENTION

Cloud-native 5G SA core networks introduce multi-dimensional operational complexity not present in legacy 4G Evolved Packet Core (EPC) deployments. The 3GPP-defined 5G Service-Based Architecture (SBA) decomposes the mobile core into distinct network functions including the Access and Mobility Management Function (AMF), Session Management Function (SMF), User Plane Function (UPF), Policy Control Function (PCF), Network Repository Function (NRF), Unified Data Management (UDM), Authentication Server Function (AUSF), and others, each communicating via service-based interfaces (SBI) using HTTP/2 and JSON-based APIs.

A single physical or cloud-native infrastructure may simultaneously serve multiple network slice instances, each identified by a Single Network Slice Selection Assistance Information (S-NSSAI) value, and each composed of a distinct set of NF instances that may be wholly dedicated or partially shared across slices. This architectural characteristic gives rise to a fault isolation problem unique to 5G SA: an anomaly detected in a slice KPI (e.g., degraded PDU session establishment success rate for eMBB slice S-NSSAI 1:1) may originate from (a) a fault in an NF shared across multiple slices, in which case multiple slice KPIs would be jointly affected; or (b) a fault local to slice-specific configuration, policy, or session state, in which case only that slice's KPIs would degrade while other slices sharing the same NF continue to operate normally.

Current network operations center (NOC) toolchains rely predominantly on threshold-based alerting and correlation engines that identify co-occurring anomalies but are fundamentally incapable of distinguishing causal relationships from coincidental co-occurrence. Existing approaches suffer from the following deficiencies:

- Correlation-based alert systems generate high volumes of redundant alerts during fault events, overwhelming operations staff and obscuring root cause identification.
- Static topology maps reflect physical or logical connectivity but do not capture dynamic, time-varying causal dependencies between NFs under varying traffic conditions.
- Manual RCA processes in virtualized 5G cores require specialized expertise across radio, transport, and core domains and can extend fault mean-time-to-repair (MTTR) from minutes to hours.
- Current observability platforms lack mechanisms to ingest, correlate, and causally analyze heterogeneous telemetry streams including PFCP/N4 session-level signals, SBI inter-NF call traces, Prometheus-exported metrics, and structured log events within a unified causal inference framework.

Prior art in causal-graph-based root cause analysis — including CauseInfer (Chen et al., 2014), MicroCause (Meng et al., 2020), and MicroDiag (Wang et al., 2021) — demonstrates the utility of causal discovery algorithms such as the PC algorithm for identifying root causes in microservice systems. However, these systems are not designed for, and do not address, the structural properties of 5G SA core networks: none incorporate S-NSSAI slice identity into graph construction; none model PFCP session bindings on the N4 interface as causal edges; none use SBI HTTP/2 call sequences as structural priors for causal graph topology; and none implement the NF-layer versus slice-layer root cause disambiguation that is operationally essential for 5G multi-slice deployments.

There remains a clear and unmet need in the art for a causal root cause analysis system that is natively aware of 5G SA core network topology, incorporating slice subgraph structure, SBI service mesh dependencies, and PFCP session state into a unified bi-level causal DAG, enabling precise and automated fault isolation across NF and slice domains.

3. SUMMARY OF THE INVENTION

In one aspect, the invention provides a computer-implemented method for root cause analysis in a cloud-native 5G standalone (SA) core network, comprising: maintaining a bi-level causal DAG with NF-layer nodes and slice subgraph projections; receiving multi-source telemetry including PFCP N4 session metrics and SBI operational data; performing conditional independence-based causal discovery constrained by the 5G topology as a structural prior; and isolating faults as either NF-layer or slice-layer root causes based on causal attribution score distribution across the bi-level graph.

In a further aspect, the method is extended to a four-domain hierarchical graph spanning RAN, transport, core, and cloud infrastructure domains, enabling cross-domain root cause propagation at scale.

In a still further aspect, the method is extended with a closed-loop remediation action mapper that automatically selects and executes remediation actions — including NF scale-out, PFCP session re-establishment, and PCF policy tightening — and verifies their effectiveness via post-remediation causal re-evaluation.

In another aspect, the invention provides a system comprising one or more processors and memory implementing the foregoing method, and a non-transitory computer-readable medium storing instructions implementing the foregoing method.

Key advantages of the present invention include: reduction in mean-time-to-repair (MTTR) for 5G core faults; elimination of alert storm confusion through causal filtering; NF-layer versus slice-layer fault disambiguation not present in any prior art; applicability across multi-vendor Open RAN and 5G SA deployments; and extensibility to next-generation 6G intent-driven network architectures.

4. BRIEF DESCRIPTION OF THE DRAWINGS

The following figures are incorporated herein and form part of this provisional patent application.

- Figure 1A — Bi-Level Causal DAG: NF Layer (Level 1). Nodes represent 5G Core NFs (AMF, SMF, UPF, PCF, NRF, AUSF, UDM). Directed edges represent causal dependencies inferred from SBI HTTP/2 call sequences captured via the 5G core service mesh. Edge weights correspond to conditional mutual information scores. The causal attribution score for each node is shown as a heatmap overlay; in the illustrated embodiment, the SMF node exhibits the highest attribution score, indicating it as the NF-layer root cause candidate.
- Figure 1B — Bi-Level Causal DAG: Slice Subgraphs (Level 2). Three slice subgraphs corresponding to S-NSSAI values 1:1 (eMBB), 1:2 (URLLC), and 1:3 (mIoT). Shared NF nodes (AMF and NRF) appear across multiple subgraphs with a visual indicator. Causal attribution scores shown per-subgraph enable identification of whether anomalous attribution concentrates across all subgraphs (NF-layer fault) or within a single subgraph (slice-layer fault).
- Figure 2 — Four-Domain Hierarchical Graph. Four horizontal layers: (a) RAN Domain comprising gNB and CU/DU-split nodes with radio KPIs; (b) Transport Domain comprising N2, N3, N9 backhaul link nodes; (c) Core Domain comprising the bi-level DAG; (d) Cloud Infrastructure comprising VM, container, and Kubernetes pod nodes. Cross-domain causal edges shown as dashed vertical arrows. In the illustrated embodiment, a Kubernetes SMF pod CPU throttle propagates upward to manifest as an NF-layer SMF fault and subsequently as degraded eMBB PDU session establishment.
- Figure 3 — Closed-Loop Remediation Flow. Sequence flow with five stages: (1) Root Cause Isolation; (2) Policy Store Lookup; (3) Action Selection; (4) Action Execution with three branches (NF scale-out, PFCP session re-establishment, PCF policy tightening); (5) Effectiveness Verification with feedback loop back to stage 1.
- Figure 4 — System Architecture Overview. Five functional subsystems (MTIE, CIE, DCGM, RCSM, FRG) and their interfaces with 5G Core Network Functions and output destinations.
- Figure 5 — Root Cause Scoring Module (RCSM) Algorithm Flowchart including decision logic and re-scoring feedback loops.

5. DETAILED DESCRIPTION OF THE INVENTION

5.1 Key Term Definitions

The following definitions apply throughout this specification and the claims. These definitions satisfy the written description requirement under 35 U.S.C. §112.

5.1.1 Network Slice and S-NSSAI

Network Slice: A logical end-to-end network defined per 3GPP TS 23.501, instantiated on a shared physical or virtual infrastructure and customized to serve a particular service type. A network slice spans RAN, transport, and core network domains and is identified by a Single Network Slice Selection Assistance Information (S-NSSAI) value.

S-NSSAI: A two-component identifier defined per 3GPP TS 23.003 comprising a Slice/Service Type (SST) field of 8 bits and an optional Slice Differentiator (SD) field of 24 bits. S-NSSAI values in the examples herein are expressed as SST:SD (e.g., 1:1 for eMBB, 1:2 for URLLC, 1:3 for mIoT).

Slice Subgraph: As used herein, a slice subgraph is the projection of the Level 1 NF-layer causal DAG onto the subset of NF nodes that participate in serving a specific network slice instance identified by an S-NSSAI value. A slice subgraph is a subgraph of the bi-level causal DAG, not an independent graph; nodes shared across multiple slice subgraphs retain a single identity in the Level 1 graph and appear as shared projection nodes in multiple Level 2 slice subgraphs.

5.1.2 Service-Based Interface (SBI) and SBI Topology Prior

Service-Based Interface (SBI): Per 3GPP TS 23.501 and TS 29.500, NFs expose their capabilities as services consumable by other NFs via standardized RESTful HTTP/2 APIs. SBI call sequences refers to the temporal ordering and causal dependencies among HTTP/2 API calls exchanged between NF service producers and consumers, as captured by a service mesh or equivalent telemetry collection mechanism.

SBI Topology Prior: A structural prior constraint applied to the causal discovery algorithm, limiting candidate causal edges in the bi-level DAG to pairs of nodes that exhibit known SBI producer-consumer relationships as defined in 3GPP specifications or as observed in the service mesh call graph. This constraint reduces the search space of the causal discovery procedure and encodes domain knowledge to improve causal graph accuracy.

5.1.3 PFCP and N4 Interface

PFCP (Packet Forwarding Control Protocol): A protocol defined in 3GPP TS 29.244 and IETF RFC 8805 for communication between the SMF and UPF via the N4 interface. PFCP messages establish, modify, and delete Packet Detection Rules (PDRs), Forwarding Action Rules (FARs), and Usage Reporting Rules (URRs) for individual PDU sessions. PFCP session statistics — including packet counts, byte counts, session establishment latency, and rule application failures — constitute one source of telemetry ingested by the system.

PFCP Session Binding: The association between a specific PDU session (identified by PDU session ID, SUPI, and S-NSSAI) and a PFCP session established between a specific SMF instance and a specific UPF instance. PFCP session bindings are used as structural prior constraints in the bi-level causal DAG, ensuring that causal edges involving UPF nodes are conditioned on the specific SMF-UPF PFCP binding rather than aggregate UPF metrics.

5.1.4 Causal DAG, Attribution Score, NF-Layer vs. Slice-Layer Root Cause

Causal Directed Acyclic Graph (Causal DAG): A directed acyclic graph G = (V, E) in which each node v in V represents a time-series metric variable associated with a network entity, and each directed edge (u, v) in E represents a causal relationship from u to v after conditioning on all other variables in the graph. The causal DAG is constructed from observational telemetry data using a

conditional independence-based causal discovery algorithm constrained by the 5G topology structural prior.

Causal Attribution Score: A numerical score assigned to each node quantifying the degree to which that node's metric time series is a root cause of an observed anomaly. Computed using PageRank-based causal scoring, structural causal model residual scoring, or intervention-based attribution scoring. A node with a causal attribution score exceeding a configurable threshold is identified as a root cause candidate.

NF-Layer Root Cause: A root cause originating in an NF shared across multiple slice subgraphs, such that causal attribution scores concentrate on a Level 1 NF node and propagate to affect multiple Level 2 slice subgraphs simultaneously.

Slice-Layer Root Cause: A root cause originating in slice-specific configuration, policy state, or session state, such that causal attribution scores concentrate on nodes within a single Level 2 slice subgraph while peer slices sharing the same NF remain unaffected.

5.2 System Architecture Overview

Referring to Figure 4, the fault isolation system (hereinafter 'the System') operates as a software-defined analytical platform deployable as a cloud-native microservice within or adjacent to the 5G core network infrastructure. The System interfaces with 5G network functions through standard telemetry export mechanisms and does not require modification to 3GPP-compliant NF implementations.

The System comprises five functional subsystems: (1) the Multi-Modal Telemetry Ingestion Engine (MTIE); (2) the Causal Inference Engine (CIE); (3) the Dynamic Causal Graph Manager (DCGM); (4) the Root Cause Scoring Module (RCSM); and (5) the Fault Report Generator (FRG). Each subsystem is described in detail in the following sections.

5.3 Multi-Modal Telemetry Ingestion Engine (MTIE)

The MTIE continuously collects and normalizes heterogeneous telemetry signals from 5G core network functions in real time. The MTIE handles four primary telemetry modalities:

(a) PFCP/N4 Interface Signals: The PFCP protocol operates over the N4 reference point between the SMF and UPF. The MTIE parses PFCP message sequences to extract session-level events including cause codes, rule application failures, and forwarding anomalies. These signals provide ground-truth visibility into user plane data path integrity.

(b) SBI HTTP/2 Inter-NF Call Traces: The MTIE captures distributed trace context propagation compatible with W3C TraceContext and OpenTelemetry standards across SBI calls including N11 (AMF-SMF), N7 (SMF-PCF), N8 (AMF-UDM), and N12 (AMF-AUSF) interfaces. Call latency distributions, error rates, retry patterns, and timeout events constitute the SBI telemetry feature set.

(c) Prometheus-Exported Performance Metrics: Standard Prometheus scrape endpoints exposed by 5G NF implementations provide time-series performance counters including registration success/failure rates, PDU session establishment rates, handover success rates, NF service availability, and resource utilization metrics. The MTIE ingests these at configurable scrape intervals and aligns them to a unified timestamp domain.

(d) Structured Log Events: Structured logs emitted by NFs in JSON or key-value format, including 3GPP-defined procedure logs, are ingested and parsed. Log-level event types including authentication failures, policy mismatches, NF deregistration events, and overload conditions are extracted and normalized.

All ingested telemetry is timestamped, tagged with source NF identity and S-NSSAI slice identity, and streamed into a unified event bus accessible by the Causal Inference Engine.

5.4 Causal Inference Engine (CIE)

The CIE applies causal inference methodology to the normalized telemetry event stream to identify directed causal relationships between NF-level events. The CIE operates on sliding temporal windows of configurable duration and employs one or more causal discovery algorithms:

(a) PC Algorithm (Primary): The PC algorithm (Spirtes, Glymour, Scheines, 1993) performs conditional independence skeleton discovery followed by edge orientation. The implementation uses partial correlation-based conditional independence tests parameterized with a configurable significance threshold alpha. The 5G SBI topology is encoded as a structural prior constraining the skeleton discovery phase to candidate edges consistent with known 3GPP NF producer-consumer relationships.

(b) PCMCI Algorithm (Time-Series Extension): The PCMCI algorithm (Runge et al., 2019, Science Advances) combines PC algorithm skeleton discovery with Momentary Conditional Independence (MCI) tests optimized for autocorrelated time series. PCMCI accepts a maximum time lag parameter tau_max controlling the window over which lagged causal dependencies are tested, enabling discovery of time-lagged causal edges that represent propagation delays between NF-layer and slice-layer fault manifestation. PCMCI is particularly well-suited for network telemetry time series, which exhibit strong autocorrelation and non-stationarity.

(c) Granger Causality (Optional): Granger causality testing applied to time-series telemetry signals identifies whether past values of one NF's metrics statistically predict future values of another NF's metrics beyond what is predictable from the latter's own history alone.

The CIE outputs a set of directed causal edges with associated confidence weights and estimated temporal lag values representing the propagation delay between cause and effect across NF boundaries. The CIE is constrained by the SBI Topology Prior and PFCP Session Binding prior as defined in Section 5.1.

5.5 Dynamic Causal Graph Manager (DCGM)

The DCGM maintains the bi-level causal DAG as a continuously updated weighted directed graph G = (V, E). The bi-level structure comprises:

Level 1 — NF Layer: Nodes represent individual 5G Core NFs (AMF, SMF, UPF, PCF, NRF, AUSF, UDM). Directed edges are derived from SBI HTTP/2 call sequences and updated by the CIE output. Each edge carries a conditional mutual information score and temporal lag value.

Level 2 — Slice Subgraphs: Per-S-NSSAI slice subgraphs are constructed as projections of the Level 1 NF nodes onto the subset of NFs participating in each slice instance. Shared NFs (e.g., a single AMF serving multiple slices) appear as shared projection nodes in multiple Level 2 subgraphs, retaining their single Level 1 identity. Slice-specific NF instances appear as non-shared nodes within their respective subgraphs.

The DCGM maintains a rolling historical buffer of graph snapshots enabling retrospective causal analysis and correlation of fault events with change management records. Graph snapshots are indexed by timestamp and annotated with active S-NSSAI set, active NF instance set, and Kubernetes pod mapping.

5.6 Root Cause Scoring Module (RCSM)

Upon detection of an anomalous telemetry pattern meeting configurable threshold criteria, the RCSM is invoked to identify the root-cause NF. The RCSM applies graph-theoretic scoring methods to the current causal graph state, as illustrated in Figure 5:

(a) In-degree and out-degree centrality analysis identifies NFs that are causal sources (high out-degree) versus effects (high in-degree) in the fault propagation subgraph.

(b) Temporal ordering analysis identifies the NF exhibiting anomalous behavior earliest in the causal chain, weighted by the temporal lag estimates from the CIE.

(c) Confidence-weighted path analysis identifies the highest-confidence causal path from the hypothesized root cause to all observed anomalous NFs.

(d) NF-Layer vs. Slice-Layer Disambiguation: The RCSM evaluates causal attribution score distribution across Level 1 nodes and Level 2 slice subgraph nodes. If attribution scores concentrate on a Level 1 NF node shared across multiple Level 2 slice subgraphs, an NF-layer root cause is declared. If attribution scores concentrate within a single Level 2 slice subgraph while peer slices remain unaffected, a slice-layer root cause is declared. This disambiguation mechanism is novel and not present in any identified prior art.

The RCSM outputs a ranked list of candidate root-cause NFs with associated confidence scores, supporting evidence identifiers, causal path descriptions, and the NF-layer versus slice-layer root cause classification.

5.7 Fault Report Generator (FRG)

The FRG synthesizes RCSM outputs into a structured fault isolation report. The report includes: the identified root-cause NF with confidence score; the causal propagation path; supporting telemetry evidence from each modality; a timeline of fault propagation events; the estimated fault onset time; the NF-layer or slice-layer root cause classification; the affected S-NSSAI identifiers; the implicated N4 PFCP session identifier or SBI service operation name; and recommended remediation actions mapped to 3GPP TS 32.111 fault management categories.

The fault report is made available via a northbound REST API compatible with OSS/BSS integration and may optionally trigger automated remediation workflows through integration with the O-RAN Non-RT RIC or SMO framework.

5.8 Four-Domain Hierarchical Graph (Claim 2 Embodiment)

In the embodiment of Claim 2, the bi-level causal DAG is organized within a four-domain hierarchical graph as illustrated in Figure 2, comprising:

- RAN Domain Layer: gNB and CU/DU-split component nodes annotated with radio KPIs including PRB utilization and PDCP retransmission rate.
- Transport Domain Layer: N2, N3, and N9 backhaul link nodes annotated with latency and jitter metrics.
- Core Domain Layer: The bi-level NF and slice subgraph structure described above.
- Cloud Infrastructure Layer: VM, container, and Kubernetes pod nodes mapped to each NF instance, annotated with CPU throttle events, memory pressure metrics, and node eviction signals.

Causal discovery is performed within each domain layer independently using domain-appropriate telemetry time granularities. Cross-domain causal edges are inferred by testing conditional independence between domain boundary metrics, enabling root cause isolation to propagate across domain boundaries without requiring a single monolithic causal graph over all domains.

5.9 Closed-Loop Remediation Action Mapper (Claim 3 Embodiment)

In the embodiment of Claim 3, the system maintains a remediation action policy store as illustrated in Figure 3. Upon isolation of a root cause, the system automatically selects a remediation action from the policy store based on root cause type and target entity, executes the action via control-plane API calls to the 5G core orchestration layer, and verifies remediation effectiveness by re-evaluating causal attribution scores over a post-remediation observation window.

Supported remediation actions include: NF instance horizontal scale-out via cloud-native orchestrator API; PFCP session re-establishment on an alternate UPF instance; slice admission control policy tightening via PCF Npcf_PolicyAuthorization service operation; and traffic steering update via SMF N4 PFCP Modification Request.

5.10 Extensibility to 6G and Intent-Based Networks

The present invention is designed with extensibility to next-generation 6G network architectures. In intent-based 6G networks, the System may be extended to monitor intent fulfillment by correlating causal graph states with declared network intents, identifying causal chains that led to intent violations, and providing causal explanations for autonomous network decisions.

6. CLAIMS

Note: Claims in a provisional patent application are optional and not examined. The following claims are included to establish the scope of the invention for priority purposes and will be refined in the non-provisional application.

Claim 1 — Independent Method Claim: Slice-Topology-Aware Causal DAG Construction and NF/Slice Root Cause Isolation

1. A computer-implemented method for root cause analysis in a cloud-native 5G standalone (SA) core network, the method comprising:

maintaining, by a monitoring system, a bi-level causal directed acyclic graph (DAG) representing a 5G core network topology, wherein:

a first level comprises nodes corresponding to individual 5G Network Functions (NFs) including at least an Access and Mobility Management Function (AMF), Session Management Function (SMF), User Plane Function (UPF), and Policy Control Function (PCF), with directed edges derived from Service-Based Interface (SBI) HTTP/2 call sequences captured via a 5G core service mesh; and

a second level comprises slice subgraphs, each slice subgraph corresponding to a distinct network slice instance identified by an S-NSSAI value, wherein nodes in each slice subgraph are projections of first-level NF nodes, and a single NF node may be shared across multiple slice subgraphs to represent slice-multiplexed NF deployments;

receiving, in real time, telemetry streams comprising at least:

NF-layer performance metrics collected via 3GPP Nnf service operations;

PFCP session-level statistics from N4 interface sessions between SMF and UPF instances; and

per-slice key performance indicator (KPI) measurements including at least PDU session establishment success rate (PSESR), user-plane latency, and packet loss ratio, each measured per S-NSSAI;

constructing, using a conditional independence-based causal discovery algorithm executed over a sliding time window applied to the received telemetry, a causal graph capturing temporal and statistical dependencies among NF-layer and slice-layer metrics, wherein the causal discovery algorithm is parameterized to respect the bi-level topology as a structural prior, constraining candidate causal edges to pairs of nodes that share an SBI producer-consumer dependency or a PFCP session binding;

isolating a root cause of a detected network anomaly by:

identifying, at the first level, whether causal attribution scores concentrate on an NF node shared across multiple slice subgraphs, thereby indicating an NF-layer root cause; or

identifying, at the second level, whether causal attribution scores concentrate on nodes within a single slice subgraph while peer slices sharing the same NF node remain unaffected, thereby indicating a slice-layer root cause; and

generating a root cause report specifying: the identified root cause type; its causal attribution score; the affected S-NSSAI identifiers; and the implicated N4 PFCP session identifier or SBI service operation name.

Claim 2 — Dependent Method Claim: Hierarchical Four-Domain Graph

2. The method of Claim 1, further comprising:

organizing the bi-level causal DAG within a four-domain hierarchical graph comprising: a RAN domain layer comprising gNB and CU/DU-split component nodes annotated with radio key performance indicators including Physical Resource Block (PRB) utilization and Packet Data Convergence Protocol (PDCP) retransmission rate; a transport domain layer comprising N2, N3, and N9 backhaul link nodes annotated with latency and jitter metrics; a core domain layer comprising the bi-level NF and slice subgraph structure of Claim 1; and a cloud infrastructure layer comprising virtual machine, container, and Kubernetes pod nodes mapped to each NF instance and annotated with CPU throttle events, memory pressure metrics, and node eviction signals;

wherein causal discovery is performed within each domain layer independently using domain-appropriate telemetry time granularities; and

inferring cross-domain causal edges by testing conditional independence between domain boundary metrics, enabling root cause isolation to propagate across domain boundaries without requiring a single monolithic causal graph over all domains.

Claim 3 — Dependent Method Claim: Closed-Loop Remediation Action Mapper

3. The method of Claim 1, further comprising:

maintaining a remediation action policy store associating causal root cause signatures with candidate remediation actions, wherein each policy entry specifies: a root cause type selected from the group consisting of NF-layer fault, slice-layer congestion, PFCP session binding failure, SBI timeout cascade, and cloud infrastructure resource exhaustion; a target entity expressed as at least one of an NF instance identifier, a slice S-NSSAI value, or a Kubernetes pod label selector; and a remediation action selected from the group consisting of NF instance horizontal scale-out via a cloud-native orchestrator API, PFCP session re-establishment on an alternate UPF instance, slice admission control policy tightening via a PCF Npcf_PolicyAuthorization service operation, and traffic steering update via an SMF N4 PFCP Modification Request;

upon isolation of a root cause pursuant to Claim 1, automatically selecting a remediation action from the policy store based on the root cause type and the target entity;

executing the selected remediation action by issuing corresponding control-plane API calls to the 5G core orchestration layer; and

verifying remediation effectiveness by re-evaluating causal attribution scores over a post-remediation observation window, and generating an alert if causal attribution scores on the identified root cause node do not decrease below a configurable threshold within the observation window.

Claim 4 — Independent System/Apparatus Claim: System with PCMCI

4. A system for root cause analysis in a cloud-native 5G standalone (SA) core network, the system comprising:

one or more processors; and

memory storing instructions that, when executed by the one or more processors, cause the system to perform operations comprising:

maintaining a bi-level causal directed acyclic graph (DAG) wherein a first level comprises NF nodes with SBI-derived causal edges and a second level comprises per-S-NSSAI slice subgraphs as projections of the first-level NF nodes, as described in Claim 1;

receiving real-time telemetry comprising NF-layer SBI metrics, N4 PFCP session statistics, and per-S-NSSAI slice KPIs, as described in Claim 1;

executing the PCMCI causal discovery algorithm over a sliding time window applied to the received telemetry, wherein the PCMCI algorithm applies a conditional independence skeleton discovery phase constrained by the 5G topology structural prior of Claim 1; applies Momentary Conditional Independence (MCI) tests to each candidate causal edge over a configurable maximum time lag parameter tau_max; and produces a time-lagged causal DAG in which each directed edge is annotated with a time lag value representing the causal delay between the source and target metric variables;

isolating a root cause of a detected network anomaly by performing the NF-layer versus slice-layer attribution analysis described in Claim 1 on the PCMCI-derived time-lagged causal DAG; and

generating a root cause report as described in Claim 1, additionally specifying the time lag value of the highest-scoring causal edge implicating the identified root cause node.

Claim 5 — Non-Transitory Computer-Readable Medium

5. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the processors to perform the method of Claim 1.

Claim 6 — O-RAN Integration

6. The method of Claim 1, wherein the root cause report is made available via a northbound REST API, and wherein the method further comprises triggering an automated remediation workflow based on the root cause report through integration with an O-RAN Non-Real-Time RAN Intelligent Controller (Non-RT RIC) or Service Management and Orchestration (SMO) framework.

7. PRIOR ART DIFFERENTIATION

The following table summarizes how the claims of the present invention differentiate from the identified prior art. All features in the Causal5G column represent novel claim elements not taught or suggested by the listed prior art references.

Feature	CauseInfer (2014)	MicroCause (2020)	MicroDiag (2021)	Causal5G (This Invention)
S-NSSAI slice identity in graph construction	No	No	No	YES
PFCP N4 session bindings as causal edges	No	No	No	YES
SBI HTTP/2 topology as structural prior	No	No	No	YES
NF-layer vs. slice-layer disambiguation	No	No	No	YES
Bi-level DAG (NF + per-S-NSSAI subgraphs)	No	No	No	YES
PCMCI with time-lagged 5G telemetry	No	No	No	YES
Closed-loop remediation with causal re-eval	No	No	No	YES
Designed for containerized 5G SA core	No	No	No	YES

8. ABSTRACT

A method, system, and non-transitory computer-readable medium for automated real-time fault isolation and root cause analysis in cloud-native 5G Standalone (SA) core networks. The invention maintains a bi-level causal directed acyclic graph (DAG) wherein a first level represents 5G Network Functions (AMF, SMF, UPF, PCF) with directed edges derived from Service-Based Interface (SBI) HTTP/2 call sequences, and a second level comprises per-S-NSSAI slice subgraphs as projections of the NF-layer graph. The system continuously ingests multi-modal telemetry including PFCP/N4 session statistics, SBI inter-NF call traces, Prometheus metrics, and structured logs, and applies conditional independence-based causal discovery constrained by the 5G SBI topology and PFCP session binding structural priors. Upon fault detection, causal attribution scores are computed across the bi-level DAG to isolate root causes as either NF-layer faults (shared NF affected, multiple slices degraded) or slice-layer faults (single slice degraded, shared NF unaffected). A closed-loop remediation mapper automatically selects and executes remediation actions and verifies their effectiveness via post-remediation causal re-evaluation. The invention addresses the fundamental deficiency of correlation-based fault management toolchains in disaggregated 5G core deployments and is the first system to incorporate S-NSSAI slice topology, PFCP session bindings, and SBI HTTP/2 structural priors into a unified causal DAG for 5G SA core fault isolation.

9. INVENTOR DECLARATION

I hereby declare that:

1. I am the sole original inventor of the subject matter claimed in this provisional patent application.
2. I have reviewed and understand the contents of this application.
3. I acknowledge my duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 C.F.R. §1.56.
4. I have not made this invention available to the public through any publication, public use, sale, or other disclosure prior to the filing date of this application, except as protected under 35 U.S.C. §102(b)(1).
5. I am a micro entity as defined under 37 C.F.R. §1.29.

Signature:	G. Krish
Printed Name:	Krishna Kumar Gattupalli
Date:	04/24/2026
Address:	Celina, TX 75009, United States